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Inventor(s)

KAMEZAKI; Mao et al.

FLUSH WATER TANK DEVICE, AND FLUSH TOILET APPARATUS INCLUDING THE SAME

Abstract

The present invention provides a flush water tank device, including a flush water tank main body including a first tank part and a second tank part for storing flush water, a rim spouting switch mechanism that switches discharge and stop of flush water in the first tank part, and a jet spouting discharge valve that switches discharge and stop of flush water in the second tank part, the first tank part and the second tank part are configured to have different cross-sectional areas depending on a height, and a ratio of the cross-sectional area of the first tank part to that of the second tank part at a first height is different from a ratio of the cross-sectional area at a second height that is lower than the first height.

Inventors: KAMEZAKI; Mao (Fukuoka, JP), KASHIMURA; Hideaki (Fukuoka, JP), MATSUDA; Haruki (Fukuoka, JP), SHINOHARA; Koki (Fukuoka, JP), SUENAGA; Mitsuhiro (Fukuoka, JP)

Applicant: TOTO LTD. (Fukuoka, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of Japanese Patent Application No. 2024-029551 filed on Feb. 29, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a flush water tank device, particularly to a flush water tank device attached to a flush toilet main body for use to supply flush water, and a flush toilet apparatus including the flush water tank device.

Description of the Related Art

[0003] In Chinese Patent Laid-Open No. CN103967088 (Patent Literature 1), a toilet water tank device is described. In this toilet water tank device, a partition formed by a wall surface vertical to the interior of a water tank is provided, and this partition constitutes a first water tank and a second water tank. The first water tank and the second water tank include a first discharge valve and a second discharge valve, respectively, and are configured so that flush water in the first water tank and flush water in the second water tank are supplied to a flush toilet.

[0004] In the toilet water tank device according to Patent Literature 1, when the first discharge valve is opened, flush water is spouted by a hydraulic head pressure of flush water in the first water tank, and when the second discharge valve is opened, flush water is discharged by a hydraulic head pressure of flush water in the second water tank. However, in the toilet water tank device described in Patent Literature 1, the water tank is simply partitioned to the first water tank and the second water tank, and hence there is a problem that a volume of each water tank and the hydraulic head pressure of flush water in the water tank cannot be sufficiently utilized for toilet flushing.

[0005] Accordingly, an object of the present invention is to provide a flush water tank device capable of sufficiently utilizing a volume in a flush water tank and a hydraulic head pressure of flush water in the flush water tank for toilet flushing, and a flush toilet apparatus including the flush water tank device.

SUMMARY OF THE INVENTION

[0006] To achieve the above object, the present invention provides a flush water tank device which is attached to a flush toilet main body for use to supply flush water, including a flush water tank main body including a first tank part and a second tank part for storing flush water supplied to the flush toilet main body, a rim spouting switch mechanism that switches discharge and stop of flush water in the first tank part so that the flush water in the first tank part is spouted from a rim spout port of the flush toilet main body, and a jet spouting discharge valve that switches discharge and stop of flush water in the second tank part so that the flush water in the second tank part is spouted from a jet spout port of the flush toilet main body, wherein the first tank part and the second tank part are configured to have different cross-sectional areas depending on a height, and a ratio of the cross-sectional area of the first tank part to the cross-sectional area of the second tank part at a first height is different from a ratio of the cross-sectional area at a second height that is lower than the first height.

[0007] Thus, in the present invention, the first tank part and the second tank part are configured to have different cross-sectional areas depending on the height, and the ratio of the cross-sectional area of the first tank part to the cross-sectional area of the second tank part at the first height is different from the ratio of the cross-sectional area at the second height that is lower than the first height. According to the present invention including this configuration, for example, when the

second tank part is configured so that the cross-sectional area of the second tank part is large at the first height and small at the second height, and in the initial stage of flushing, a large amount of flush water can be spouted from the jet spout port at a high hydraulic head pressure. That is, according to the present invention, it is possible to distribute the flush water in the flush water tank main body to the rim spout port and the jet spout port of the flush toilet main body in accordance with a required amount of flush water and water force. Consequently, the volume in the flush water tank main body and the hydraulic head pressure of flush water in the flush water tank main body can be fully utilized for toilet flushing.

[0008] In the present invention, preferably, the second tank part is configured to have a cross-sectional area that is equal to or more than the cross-sectional area of the first tank part at the first height.

[0009] According to the present invention including this configuration, since the second tank part is configured to have a cross-sectional area that is equal to or more than the cross-sectional area of the first tank part at the first height, in the initial stage of flushing in which a water level in the second tank part is high, an instantaneous flow rate of flush water spouted from the jet spout port of the flush toilet main body can be increased, and flushing effect can be enhanced.

[0010] In the present invention, preferably, the second tank part includes a stepped portion provided at a height between the first height and the second height, and the second tank part is configured to have a reduced cross-sectional area below the stepped portion, while the first tank part is configured to have an enlarged cross-sectional area below the stepped portion.

[0011] According to the present invention including this configuration, the second tank part is configured to have the reduced cross-sectional area below the stepped portion, while the first tank part is configured to have the enlarged cross-sectional area below the stepped portion. As a result, in the initial stage of flushing, the instantaneous flow rate from the jet spout port can be increased, and concurrently, a sufficient volume can be acquired in the first tank part at the height that is equal to or less than the height of the stepped portion, so that a sufficient amount of flush water can be allocated to the rim spout port.

[0012] In the present invention, preferably, an inner tank is disposed in the flush water tank main body, the second tank part is formed inside the inner tank, and the first tank part is formed inside the flush water tank main body and outside the inner tank.

[0013] According to the present invention including this configuration, the inner tank is disposed in the flush water tank main body, the second tank part is formed inside the inner tank, and the first tank part is formed inside the flush water tank main body and outside the inner tank. Therefore, the cross-sectional area of each of the first and second tank parts at each height can be freely set, and the amount of flush water and the hydraulic head pressure can be freely set as required.

[0014] In the present invention, preferably, the jet spouting discharge valve is configured to discharge the flush water in the second tank part so that a water level in the second tank part lowers from a predetermined initial water level to a predetermined jet dead water level, and the stepped portion is provided at a position that is lower than the jet dead water level.

[0015] According to the present invention including this configuration, since the height of the stepped portion in which the cross-sectional area of the second tank part is reduced is set to be lower than the jet dead water level, the flow rate of flush water spouted from the jet spout port can be maintained to be sufficiently high until the spouting is ended.

[0016] In the present invention, preferably, the rim spouting switch mechanism is configured to discharge the flush water in the first tank part so that the water level in the first tank part lowers from the predetermined initial water level to a predetermined rim dead water level, and the stepped portion is provided at a position that is higher than the rim dead water level.

[0017] According to the present invention including this configuration, since the stepped portion in which the cross-sectional area of the first tank part enlarges is provided at a position higher than the rim dead water level, the flush water stored below the stepped portion of the first tank part can be

spouted from the rim spout port, and a sufficient amount of flush water spouted from the rim spout port can be acquired.

[0018] In the present invention, preferably, the first tank part and the second tank part are configured so that the ratio of the cross-sectional area is reversed at the first height and the second height.

[0019] According to the present invention including this configuration, since the first tank part and the second tank part are configured so that the ratio of the cross-sectional area is reversed at the first height and the second height, the instantaneous flow rate of flush water spouted from either one of the jet spout port and the rim spout port can be increased. Concurrently, enough flush water spouted from the other port can be acquired.

[0020] Furthermore, the present invention provides a flush toilet apparatus that is flushed with flush water stored in a flush water tank, including a flush toilet main body including a bowl, a rim spout port, and a jet spout port, and the flush water tank device of the present invention that supplies flush water to the rim spout port and the jet spout port of the flush toilet main body.

[0021] According to a flush water tank device of the present invention, and a flush toilet apparatus including the flush water tank device, a volume in a flush water tank and a hydraulic head pressure of flush water in the flush water tank can be sufficiently utilized for toilet flushing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a block diagram showing an entire configuration of a flush toilet apparatus according to an embodiment of the present invention;

[0023] FIG. 2 is a top view showing a schematic configuration of the flush toilet apparatus according to the embodiment of the present invention;

[0024] FIG. 3 is a front cross-sectional view showing a schematic configuration of a flush water tank device provided in the flush toilet apparatus according to the embodiment of the present invention;

[0025] FIG. 4 is a cross-sectional view showing a structure of a ball tap built in the flush water tank device according to the embodiment of the present invention;

[0026] FIG. 5 is a cross-sectional view showing a structure of a hydraulic drive mechanism built in the flush water tank device according to the embodiment of the present invention;

[0027] FIG. 6 is a horizontal cross-sectional view of the flush water tank device according to the embodiment of the present invention at a height H1 of FIG. 3;

[0028] FIG. 7 is a horizontal cross-sectional view of the flush water tank device according to the embodiment of the present invention at a height H2 of FIG. 3; and

[0029] FIG. 8 is a time chart showing the operation of the flush toilet apparatus according to the embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0030] Next, with reference to the accompanying drawings, a flush water tank device according to an embodiment of the present invention and a flush toilet apparatus including the same will be described.

[0031] FIG. 1 is a block diagram showing an entire configuration of the flush toilet apparatus according to the embodiment of the present invention. FIG. 2 is a top view showing a schematic configuration of the flush toilet apparatus according to the embodiment of the present invention. FIG. 3 is a front cross-sectional view showing a schematic configuration of a flush water tank device provided in the flush toilet apparatus according to the embodiment of the present invention. FIG. 4 is a cross-sectional view showing a structure of a ball tap built in the flush water tank device according to the embodiment of the present invention. FIG. 5 is a cross-sectional view showing a

structure of a hydraulic drive mechanism built in the flush water tank device according to the embodiment of the present invention. FIG. 6 is a horizontal cross-sectional view of the flush water tank device at a height H1 of FIG. 3. FIG. 7 is a horizontal cross-sectional view of the flush water tank device at a height H2 of FIG. 3.

[0032] As shown in FIGS. 1 and 2, a flush toilet apparatus 1 according to the embodiment of the present invention includes a flush toilet main body 2 and a flush water tank device 4 disposed at the rear of the flush toilet main body 2. The flush toilet apparatus 1 of the present embodiment is configured to be flushed by operating a lever handle 4a provided in the flush water tank device 4 after use.

[0033] The flush toilet main body 2 includes a bowl 2a and a discharge trap conduit 2b extending from a lower part of the bowl 2a. Furthermore, a rim spout port 2c is provided in a top edge portion of the bowl 2a, and a jet spout port 2d is provided in the lower part of the bowl 2a. During toilet flushing, flush water is spouted from each of the rim spout port 2c and the jet spout port 2d at a predetermined timing, and a waste receiving surface of the bowl 2a is flushed, while waste and flush water in the bowl 2a are drained to the discharge trap conduit 2b. The waste and flush water drained to the discharge trap conduit 2b are drained through a discharge socket (not shown) to a sewage pipe (not shown).

[0034] Flush water is supplied to the flush water tank device 4 from a water supply source 6 such as a water supply, and the supplied flush water is stored in the flush water tank device 4. Furthermore, in the flush water tank device 4, a rim spouting discharge valve 8 that is a rim spouting switch mechanism, and a jet spouting discharge valve 10 are built and are configured to open and close discharge ports, respectively, provided in a bottom portion of the flush water tank device 4.

[0035] In the present embodiment, by opening the rim spouting discharge valve 8, flush water is spouted from the rim spout port 2c through a rim conduit 2e formed inside the flush toilet main body 2. Furthermore, by opening the jet spouting discharge valve 10, flush water is spouted from the jet spout port 2d through a jet conduit 2f formed inside the flush toilet main body 2.

[0036] Next, with reference to FIGS. 2 and 3, an internal structure of the flush water tank device 4 will be described.

[0037] As shown in FIGS. 2 and 3, the flush water tank device 4 includes a flush water tank main body 12, an inner tank 14 disposed inside the flush water tank main body, the rim spouting discharge valve 8 disposed in the flush water tank main body 12, the jet spouting discharge valve 10 disposed in the inner tank 14, a ball tap 16 that is a water supply valve, and a hydraulic drive mechanism 18.

[0038] The flush water tank main body 12 and the inner tank 14 are containers configured to store flush water to be supplied to the flush toilet main body 2. In the present embodiment, the flush water tank main body 12 is made of ceramic, and the inner tank 14 disposed inside the flush water tank main body 12 is made of resin. In the present embodiment, a space inside the flush water tank main body 12 and outside the inner tank 14 constitutes a first tank part 13, and a space in the inner tank 14 constitutes a second tank part 15. Therefore, in the present embodiment, the first tank part 13 and the second tank part 15 for storing flush water supplied to the flush toilet main body 2 are provided inside the flush water tank main body 12.

[0039] Furthermore, as shown in FIG. 3, a first drain port 12a and a second drain port 12b are provided on a bottom surface of the flush water tank main body 12, and in the present embodiment, these drain ports are formed in a circular shape. Here, the flush water stored in the first tank part 13 (inside the flush water tank main body 12 and outside the inner tank 14) flows through the first drain port 12a into the rim conduit 2e of the flush toilet main body 2 and is spouted from the rim spout port 2c. Furthermore, the flush water stored in the second tank part 15 (in the inner tank 14) flows through the second drain port 12b into the jet conduit 2f of the flush toilet main body 2 and is spouted from the jet spout port 2d.

[0040] The inner tank **14** is disposed on the bottom surface of the flush water tank main body **12**, and the lower part of the inner tank **14** is submerged in the flush water stored in the flush water tank main body **12**. Furthermore, a circular drain port **14a** is formed on the bottom surface of the inner tank **14**. The drain port **14a** of the inner tank **14** is disposed concentrically to match the second drain port **12b** provided in the flush water tank main body **12**. That is, in the present embodiment, the center of the circular second drain port **12b** and the circular drain port **14a** coincide in top view. Therefore, the flush water in the inner tank **14** flows into the jet conduit **2f** of the flush toilet main body **2** through the drain port **14a** of the inner tank **14** and the second drain port **12b** of the flush water tank main body **12**.

[0041] Furthermore, as shown in FIG. 3, in the present embodiment, the drain port **14a** of the inner tank **14** is composed of a drain port forming member **14b** composed separately from a main body part of the inner tank **14**. The drain port forming member **14b** is a tubular member and is water-tightly attached to the bottom surface of the inner tank **14** to form the drain port **14a** inside. Furthermore, a seat surface is provided at an upper end of the drain port forming member **14b**, and the drain port **14a** is closed when the jet spouting discharge valve **10** is seated on the seat surface. Therefore, in the present embodiment, the seat surface on which the jet spouting discharge valve **10** is seated is composed of a member separate from the inner tank **14**.

[0042] Furthermore, as shown in FIG. 2, the flush water tank main body **12** of the flush water tank device **4** is formed in a substantially rectangular shape that is long in a width direction of the flush toilet main body **2** (arrow W direction in FIG. 2) and short in a depth direction (arrow D direction in FIG. 2) in top view. Furthermore, the inner tank **14** disposed in the flush water tank main body **12** is also formed to be long in the width direction and short in the depth direction in top view and is irregularly shaped to avoid the rim spouting discharge valve **8** and the ball tap **16**.

[0043] Specifically, in the present embodiment, the ball tap **16** is disposed in a front left end portion (lower left corner in FIG. 2) in the flush water tank main body **12**, and the rim spouting discharge valve **8** is disposed in a front right end portion (lower right corner in FIG. 2). The inner tank **14** is provided with cutouts **14c** and **14d** in opposite front end portions in the width direction in top view to avoid the ball tap **16** and the rim spouting discharge valve **8**. That is, in the flush water tank main body **12**, the cutout **14c** that is U-shaped is provided in the inner tank **14** in a portion in which the rim spouting discharge valve **8** is disposed, and the cutout **14d** is provided in a portion in which the ball tap **16** is disposed, so that the valve and ball tap do not interfere with the inner tank **14**.

[0044] On the other hand, as shown in FIG. 3, a stepped portion **14e** is formed on each side of the inner tank **14** in the width direction, and the inner tank is configured to have a reduced width below the stepped portions **14e**. That is, the inner tank **14** is substantially formed in a T-shape in front view so that a horizontal cross-sectional area **A2** (FIGS. 6 and 7) is constant above and below the stepped portion **14e**. In other words, the second tank part **15**, which is an inner space of the inner tank **14**, and the first tank part **13**, which is an outer space, are configured to have different horizontal cross-sectional areas depending on the height. That is, the second tank part **15** is configured to have a reduced horizontal cross-sectional area **A2** below the stepped portion **14e**, while the first tank part **13** is configured to have an enlarged horizontal cross-sectional area **A1** below the stepped portion **14e**.

[0045] As a result, in the present embodiment, a ratio of the horizontal cross-sectional area of the first tank part **13** to that of the second tank part **15** at a first height **H1** above the stepped portion **14e** ($A1/A2$ in FIG. 6) and a ratio of the horizontal cross-sectional area of the first tank part **13** to that of the second tank part **15** at a second height **H2** below the first height **H1** and the stepped portion **14e** ($A1/A2$ in FIG. 7) are different. In the present embodiment, as shown in FIG. 6, in the first height **H1** above the stepped portion **14e**, the horizontal cross-sectional area **A2** of the second tank part **15** is larger than the horizontal cross-sectional area **A1** of the first tank part **13**. As shown in FIG. 7, in the second height **H2** below the stepped portion **14e**, the horizontal cross-sectional

area A1 of the first tank part **13** is larger than the horizontal cross-sectional area A2 of the second tank part **15**. Therefore, in the present embodiment, the ratio of the horizontal cross-sectional area of the first tank part **13** to that of the second tank part **15** is reversed at the first height H1 and the second height H2.

[0046] Furthermore, during actual use of the flush water tank device **4**, flush water is stored to a level above the stepped portion **14e** of the inner tank **14**, in the second tank part **15**. Therefore, in the present embodiment, the inner tank **14** is submerged in the flush water stored in the first tank part **13** up to the level above the stepped portion **14e**. Then, in the present embodiment, by configuring a large upper part of the second tank part **15**, a large amount of flush water having large potential energy is stored in the second tank part **15**. Consequently, the hydraulic head pressure of the flush water drained from the second tank part **15** can be increased, and the instantaneous flow rate of the flush water spouted from the jet spout port **2d** can be increased.

[0047] Furthermore, by irregularly shaping the inner tank **14** with resin, sufficient volume is acquired in the inner tank **14** while suppressing a dimension of the flush water tank device **4** in the depth direction. Additionally, the inner tank **14** made of resin can be configured with a thinner wall surface than an inner tank made of ceramic, and hence the volume in the flush water tank main body **12** can be effectively utilized.

[0048] Next, the rim spouting discharge valve **8**, which is the rim spouting switch mechanism, is a valve body disposed to open and close the first drain port **12a** provided in the flush water tank main body **12**, and the first drain port **12a** is opened by pulling the rim spouting discharge valve **8** upward. Consequently, the flush water in the first tank part **13** is drained to the rim conduit **2e** (FIG. 1) of the flush toilet main body **2** and spouted from the rim spout port **2c**. Therefore, the rim spouting discharge valve **8**, which is the rim spouting switch mechanism, switches the discharge and stop of the flush water stored in the first tank part **13** to the flush toilet main body **2**.

[0049] In the present embodiment, a bead chain **8a** (FIG. 3) coupled to the rim spouting discharge valve **8** is pulled by a user rotating and operating the lever handle **4a** provided in the flush water tank device **4**, and the rim spouting discharge valve **8** is pulled upward. As a modification, the present invention may include a configuration in which the rim spouting discharge valve **8** is pulled up based on a control signal from a remote controller (not shown) or a detection signal from a human sensor (not shown), to perform flushing.

[0050] Furthermore, in the present embodiment, the rim spouting discharge valve **8** is provided as the rim spouting switch mechanism, and a configuration other than the rim spouting discharge valve **8** may be used as the rim spouting switch mechanism. For example, a jet pump (not shown) that causes flush water stored in the first tank part **13** to flow into the flush toilet main body may be used as the rim spouting switch mechanism. In this case, a jet pump nozzle (not shown) is disposed in the first tank part **13**, and through this nozzle, part of the flush water supplied via a water supply valve such as the ball tap **16** is injected. By injecting flush water from the nozzle in the first tank part **13**, the flush water stored in the first tank part **13** can be entrained, and the flush water in the first tank part **13** can flow into the flush toilet main body by jet pump action. In addition to this jet pump (not shown), an arbitrary configuration that switches discharge and stop of flush water to the flush toilet main body **2** may be used as the rim spouting switch mechanism.

[0051] Next, the jet spouting discharge valve **10** is a valve body disposed to open and close the drain port **14a** provided in the inner tank **14**, and by pulling the jet spouting discharge valve **10** upward, the jet spouting discharge valve **10** is unseated from the seat surface of the drain port **14a**, to open the drain port **14a**. Consequently, the flush water in the second tank part **15** is drained from the drain port **14a**, flows through the second drain port **12b** into the jet conduit **2f** (FIG. 1) of the flush toilet main body **2**, and is spouted from the jet spout port **2d**. Therefore, the jet spouting discharge valve **10** switches the discharge and stop of the flush water stored in the second tank part **15** to the flush toilet main body **2**.

[0052] In the present embodiment, the jet spouting discharge valve **10** is configured to be pulled up

from the drain port **14a** by the hydraulic drive mechanism **18**. That is, the jet spouting discharge valve **10** is a valve body including a valve shaft **10a** extending upward, and the valve shaft **10a** is pulled up by the hydraulic drive mechanism **18**. Then, when pulled up to a predetermined height, the jet spouting discharge valve **10** is disconnected from the hydraulic drive mechanism **18**, and gently descends to close the drain port **14a**. The configuration of the hydraulic drive mechanism **18** will be described later, but the present invention may be applied to a flush water tank device that is not provided with the hydraulic drive mechanism **18**.

[0053] Furthermore, the ball tap **16**, which is a water supply valve, includes an inflow pipe **16a** through which flush water supplied from the water supply source **6** flows into the ball tap, and is configured to switch supply and stop of flush water to be stored in the first tank part **13** and the second tank part **15**.

[0054] Next, with reference to FIG. **4**, the configuration of the ball tap **16** will be described.

[0055] As shown in FIG. **4**, the ball tap **16** includes a main body **20** to which the inflow pipe **16a** and an outflow pipe **16b** are connected, a main valve body **20a** disposed in the main body **20**, a valve seat **20b** on which the main valve body **20a** is seated, an arm **24** rotated by a float **22**, and a pilot valve **26** moved by the rotation of the arm **24**. That is, the ball tap **16** includes the float **22** that operates in conjunction with a water level in the flush water tank device **4**, and the float **22** is configured to supply flush water to the hydraulic drive mechanism **18** when lowering to a predetermined position.

[0056] The main body **20** is a member having a lower part including a connecting portion to the inflow pipe **16a** and having one side including a connecting portion to the outflow pipe **16b**. Furthermore, the valve seat **20b** is formed inside the main body **20**, and the valve seat **20b** communicates with the outflow pipe **16b** connected to the connecting portion. Furthermore, inside the main body **20**, the main valve body **20a** is disposed to open and close the valve seat **20b** and is configured so that when the valve is opened, tap water flowing from the inflow pipe **16a** flows through the valve seat **20b**, to flow out to the outflow pipe **16b**. The outflow pipe **16b** is then connected to the hydraulic drive mechanism **18**.

[0057] The main valve body **20a** is a diaphragm valve body that is generally disc-shaped and is attached to the main body **20** so that the main valve body can be seated on and unseated from the valve seat **20b**. Furthermore, a bleed hole **20c** is provided in a peripheral edge portion of the main valve body **20a**. Furthermore, in the main valve body **20a**, a pressure chamber **20d** is formed on a side of the main valve body **20a** opposite to the valve seat **20b** (left side in FIG. **4**). That is, the pressure chamber **20d** is defined by an inner wall surface of the main body **20** and the main valve body **20a**, and when the pressure in the pressure chamber **20d** increases, the main valve body **20a** is pressed against the valve seat **20b** and seated on the valve seat **20b** by this pressure.

[0058] Furthermore, a pressure passage **20e** extends upward to communicate with the pressure chamber **20d** provided in the main body **20**, and a pilot valve port **26a** is provided at an upper end of the pressure passage **20e**. The pilot valve port **26a** is opened upward and is configured to be opened and closed by the pilot valve **26**.

[0059] On one hand, the float **22** is supported by the arm **24**, and the arm **24** is rotatably supported by a support shaft **24a**. Furthermore, the pilot valve **26** is coupled to the arm **24**, and the pilot valve **26** is configured to move in an up-down direction together with the rotation of the arm **24**. In the present embodiment, the float **22** is disposed in the second tank part **15** and moved vertically depending on the water level in the second tank part **15**. Consequently, when the water level in the second tank part **15** is raised to or above a predetermined water level, the float **22** is pushed upward, and the pilot valve **26** is accordingly moved downward and seated on the pilot valve port **26a**, to close the pilot valve port. On the other hand, when the flush water in the second tank part **15** is discharged to lower the water level, the float **22** moves downward, and the pilot valve **26** moves upward, to open the pilot valve port **26a**. Consequently, when toilet flushing is on standby in a state in which the water level in the flush water tank main body **12** is higher than the

predetermined water level, the pilot valve port **26a** of the main body **20** is closed.

[0060] Furthermore, tap water flowing through the inflow pipe **16a** into the main body **20** flows into an annular space around the valve seat **20b** and flows from here through the bleed hole **20c** of the main valve body **20a** into the pressure chamber **20d**. In this state in which the pilot valve port **26a** is closed by the pilot valve **26**, there is no path for tap water flowing from the bleed hole **20c** into the pressure chamber **20d** to flow out, and the pressure in the pressure chamber **20d** rises. When the pressure in the pressure chamber **20d** rises in this way, the main valve body **20a** is pressed toward the valve seat **20b** (rightward in FIG. 4) by this pressure, and the valve seat **20b** is closed by the main valve body **20a**.

[0061] In contrast, when the rim spouting discharge valve **8** is opened by the flushing operation and the water level in the first tank part **13** becomes lower than the predetermined water level, the float **22** descends, and the pilot valve **26** moves upward, to open the pilot valve port **26a**. When the pilot valve port **26a** is opened, water in the pressure chamber **20d** flows out of the pilot valve port **26a** and the pressure in the pressure chamber **20d** decreases. Consequently, the main valve body **20a** is moved to be pulled away from the valve seat **20b** (leftward in FIG. 4), to open the valve seat **20b**. Consequently, in this state in which the pilot valve port **26a** is opened, the pressure in the pressure chamber **20d** does not increase, so that the valve seat **20b** is opened.

[0062] Next, with reference to FIG. 5, the configuration of the hydraulic drive mechanism **18** will be described.

[0063] The hydraulic drive mechanism **18** is configured to drive the jet spouting discharge valve **10** using a water supply pressure of the flush water supplied from the water supply to the flush water tank device **4**. Specifically, the hydraulic drive mechanism **18** includes a cylinder **18a** into which water supplied from the ball tap **16** flows, a piston **18b** slidably disposed in the cylinder **18a**, and a rod **28** that protrudes from a lower end of the cylinder **18a** to drive the jet spouting discharge valve **10**. Furthermore, a spring **18c** is disposed inside the cylinder **18a**, to bias the piston **18b** downward, and a packing is attached to the piston **18b**, to acquire watertightness between an inner wall surface of the cylinder **18a** and the piston **18b**. In addition, a clutch mechanism **30** is provided at a lower end of the rod **28**, and the rod **28** and the valve shaft **10a** of the jet spouting discharge valve **10** are coupled/decoupled by the clutch mechanism **30**.

[0064] The cylinder **18a** is a cylindrical member, has an axis oriented in a vertical direction and can slidably receive the piston **18b** inside. Furthermore, the outflow pipe **16b** extending from the ball tap **16** is connected to a lower end of the cylinder **18a**, and the flush water flowing out of the ball tap **16** flows into the cylinder **18a**. Consequently, the piston **18b** in the cylinder **18a** is pushed up against a biasing force of the spring **18c** by the water flowing into the cylinder **18a**.

[0065] In addition, an outflow hole is provided in an upper end of the cylinder **18a**, and a water supply pipe **32** is connected to the outflow hole. Therefore, when water flows into the cylinder **18a** through the outflow pipe **16b** connected to a lower part of the cylinder **18a**, the piston **18b** is pushed upward from the lower part of the cylinder **18a**. Then, when the piston **18b** is pushed up to a position above the outflow hole, the water flowing into the cylinder **18a** flows out of the outflow hole to the water supply pipe **32**.

[0066] The water supply pipe **32** is a pipe extending downward from the outflow hole provided in the upper end of the cylinder **18a**. The flush water flowing out of the cylinder **18a** flows into the second tank part **15** from a water supply port **32a** at a lower end of the water supply pipe **32**. The water supply port **32a** at the lower end of the water supply pipe **32** is located below the water surface of the flush water in the second tank part **15** on standby. Therefore, the ball tap **16** is configured to cause flush water to flow into the second tank part **15** from the water supply port **32a** provided below the water surface of the second tank part **15**.

[0067] The rod **28** is a bar-shaped member connected to a lower surface of the piston **18b** and extends to protrude downward from the cylinder **18a** via a through hole formed on a bottom surface of the cylinder **18a**. Furthermore, the valve shaft **10a** of the jet spouting discharge valve **10** is

connected to the lower end of the rod **28** via the clutch mechanism **30**, and the rod **28** couples the piston **18b** and the jet spouting discharge valve **10**. Consequently, when water flows into the cylinder **18a** to push up the piston **18b**, the rod **28** connected to the piston **18b** lifts the jet spouting discharge valve **10** upward, and the jet spouting discharge valve **10** is opened.

[0068] Furthermore, a gap is provided between the rod **28** protruding from the bottom of the cylinder **18a** and the inner wall of the through hole of the cylinder **18a**, and part of the water flowing into the cylinder **18a** flows out of this gap. The water flowing out of the gap flows into the second tank part **15**. This gap is comparatively narrow and has a large flow channel resistance. Therefore, even if water flows out of the gap, the pressure in the cylinder **18a** increases due to the water flowing through the outflow pipe **16b** into the cylinder **18a**, and the piston **18b** is pushed up against the biasing force of the spring **18c**.

[0069] Furthermore, the clutch mechanism **30** removably couples the rod **28** and the jet spouting discharge valve **10**. The clutch mechanism **30** is configured to disconnect the valve shaft **10a** of the jet spouting discharge valve **10** from the rod **28**, when the jet spouting discharge valve **10** is lifted up together with the rod **28** along a predetermined distance. While the clutch mechanism **30** is disconnected, the jet spouting discharge valve **10** is no longer linked to the movement of the piston **18b** and the rod **28**, and the jet spouting discharge valve **10** descends with decrease in water level in the second tank part **15**, to close the drain port **14a** of the inner tank **14**.

[0070] Next, newly with reference to FIG. **8**, an operation of the flush toilet apparatus **1** according to the embodiment of the present invention will be described.

[0071] FIG. **8** is a time chart showing the operation of the flush toilet apparatus **1** according to the embodiment of the present invention, and shows a state of jet spouting, a state of rim spouting, and a state of the ball tap in order from the top.

[0072] First, in a standby state for toilet flushing, the first drain port **12a** of the flush water tank main body **12** and the drain port **14a** of the inner tank **14** are closed by the rim spouting discharge valve **8** and the jet spouting discharge valve **10**, respectively. Furthermore, in the standby state, flush water is stored up to a predetermined initial water level in each of the first tank part **13** and the second tank part **15**. In the present embodiment, the initial water level of the first tank part **13** is set to a position higher than the stepped portion **14e** of the inner tank **14**. Thus, in a state in which the first tank part **13** is at the initial water level, the pilot valve port **26a** of the main body **20** (FIG. **4**) of the ball tap **16** is closed, and the valve seat **20b** is closed by the main valve body **20a**. On the other hand, in the present embodiment, the initial water level of the second tank part **15** is almost as high as the upper edge of the inner tank **14** and is at a position higher than the initial water level of the first tank part **13**.

[0073] Next, at time **t1** in FIG. **8**, when the user rotates and operates the lever handle **4a** (FIG. **3**) of the flush water tank device **4** to perform toilet flushing, the bead chain **8a** connected to the handle pulls up the rim spouting discharge valve **8**. Consequently, the rim spouting discharge valve **8** is pulled away from the first drain port **12a**, and the first drain port **12a** is opened. When the first drain port **12a** is opened, the flush water stored in the second tank part **15** flows from the first drain port **12a** into the rim conduit **2e** (FIG. **2**) and is spouted from the rim spout port **2c**. By rim spouting from the rim spout port **2c**, a circulating flow is formed on the waste receiving surface of the bowl **2a**, and the waste receiving surface is flushed.

[0074] As the flush water is drained from the first drain port **12a**, the water level in the second tank part **15** decreases. Consequently, the float **22** of the ball tap **16** lowers to open the pilot valve **26** (FIG. **4**). This decreases the pressure in the pressure chamber **20d**, to open the main valve body **20a**, and flush water is supplied through the outflow pipe **16b** to the hydraulic drive mechanism **18**.

[0075] When flush water is supplied to the hydraulic drive mechanism **18**, the flush water flowing into the cylinder **18a** (FIG. **5**) pushes up the piston **18b** against the biasing force of the spring **18c**. Consequently, the rod **28** coupled to the piston **18b** pulls up the valve shaft **10a** of the jet spouting discharge valve **10**, and the drain port **14a** of the inner tank **14** is opened. That is, the jet spouting

discharge valve **10** is driven by the water supply pressure of tap water supplied via the ball tap **16** and is opened.

[0076] At time t_2 in FIG. **8**, the drain port **14a** is opened, and the flush water stored in the second tank part **15** flows through the drain port **14a** and the second drain port **12b** into the jet conduit **2f** (FIG. **2**) and is spouted from the jet spout port **2d**. The jet spouting from the jet spout port **2d** fills the discharge trap conduit **2b** with water and induces siphon action. Due to occurrence of the siphon action, retained water and waste in the bowl **2a** are suctioned into the discharge trap conduit **2b** and drained to the sewage pipe (not shown).

[0077] Here, in a state in which the drain port **14a** is closed, the interior of the jet conduit **2f** is filled with flush water to a level of pooled water surface of the bowl **2a**, and no flush water is in a space above this pooled water surface. Therefore, in a state before the jet spouting discharge valve **10** is opened, no flush water is at an upstream end in the jet conduit **2f**.

[0078] When the jet spouting discharge valve **10** is opened, the space in which there is no flush water in the jet conduit **2f** is filled in a brief time, and the flush water can be spouted from the jet spout port **2d** at an early stage. In particular, according to the present embodiment, the cross-sectional area (cross-sectional area **A2** in FIG. **6**) of the inner tank **14** above the stepped portion **14e** is larger than the cross-sectional area (cross-sectional area **A2** in FIG. **7**) below the stepped portion, and hence the second tank part **15** includes a large volume at a high position. Consequently, a large amount of flush water having a high hydraulic head pressure can flow into the jet conduit **2f**, and the instantaneous flow rate of the flush water spouted from the jet spout port **2d** can be increased.

[0079] When the jet spouting discharge valve **10** is pulled up to the predetermined height together with the piston **18b** of the hydraulic drive mechanism **18**, the valve shaft **10a** of the jet spouting discharge valve **10** is disconnected from the rod **28** by the clutch mechanism **30** (FIG. **5**).

Consequently, the jet spouting discharge valve **10** descends toward the drain port **14a** as the water level in the second tank part **15** decreases. Then, at time t_3 in FIG. **8**, when the water level in the second tank part **15** decreases to a predetermined jet dead water level JWL (FIG. **3**), the jet spouting discharge valve **10** is seated at the drain port **14a**, and the drain port **14a** is closed. This stops the jet spouting from the jet spout port **2d**.

[0080] In the present embodiment, the jet dead water level JWL is set to a level above the stepped portion **14e** of the inner tank **14**. Consequently, flush water stored at a comparatively high position in the second tank part **15** is used for spouting from the jet spout port **2d**. Therefore, a comparatively high hydraulic head pressure acts on the flush water spouted from the jet spout port **2d**, and the jet spouting can be performed at a large flow rate with a strong water force.

Furthermore, since the horizontal cross-sectional area **A2** above the stepped portion **14e** of the inner tank **14** is large, a large amount of flush water can be acquired for jet spouting.

[0081] In a state in which the main valve body **20a** of the ball tap **16** is opened, the flush water supplied from the water supply source **6** (water supply) is supplied to the hydraulic drive mechanism **18** via the ball tap **16**, and flows through the water supply pipe **32** (FIG. **5**) connected to the cylinder **18a** into the second tank part **15**. Here, in a state in which the jet spouting discharge valve **10** is opened, the flow rate of the flush water flowing out of the drain port **14a** of the inner tank **14** is larger than the flow rate of the flush water flowing through the water supply port **32a** of the water supply pipe **32** into the second tank part **15**, and hence the water level in the second tank part **15** decreases. Then, after the jet spouting is stopped at time t_3 in FIG. **8** (after the jet spouting discharge valve **10** is closed), the flush water flowing in from the water supply pipe **32** raises the water level in the second tank part **15**.

[0082] On the other hand, since the rim spouting discharge valve **8** is still in an opened state, the flush water in the first tank part **13** flows out of the first drain port **12a**, and the water level in the first tank part **13** continues to decrease. Then, at time t_4 in FIG. **8**, when the water level in the flush water tank main body **12** decreases to a predetermined rim dead water level RWL (FIG. **3**), the rim spouting discharge valve **8** is seated on the first drain port **12a**, and the rim spouting discharge

valve **8** is closed. Consequently, the rim spouting from the rim spout port **2c** is stopped.

[0083] In the present embodiment, the rim dead water level RWL is set to below the stepped portion **14e** of the inner tank **14**. For this reason, even the flush water stored at a comparatively low position in the first tank part **13** is used in spouting from the rim spout port **2c**. Therefore, an exceptionally large hydraulic head pressure does not act on the flush water spouted from the rim spout port **2c**. However, the rim spout port **2c**, which opens above the pooled water surface of the bowl **2a**, is open to the atmosphere and can facilitate spouting even if the hydraulic head pressure of flush water is low, and the bowl **2a** can be effectively flushed. Furthermore, according to the present embodiment, in the rim spouting, the flush water stored in the first tank part **13** is utilized from the initial water level above the stepped portion **14e** to the rim dead water level RWL below the stepped portion **14e**, and hence a large amount of flush water can be acquired for the rim spouting.

[0084] Furthermore, even after the rim spouting discharge valve **8** is closed, the main valve body **20a** of the ball tap **16** is maintained in an opened state, and the flush water supplied from the water supply source **6** (water supply) therefore flows into the second tank part **15** from the water supply port **32a** of the water supply pipe **32** via the ball tap **16** and the hydraulic drive mechanism **18**.

[0085] Then, the water level in the second tank part **15** rises due to the inflow of flush water, and at time t_5 in FIG. **8** at which the second tank part **15** is full, the flush water begins to overflow from the inner tank **14**, and the flush water flows into the first tank part **13**.

[0086] As flush water flows into the first tank part **13**, the water level in the first tank part **13** begins to rise. Next, the water level in the first tank part **13** rises to a predetermined initial water level, and the float **22** of the ball tap **16** rises, to close the pilot valve **26** (FIG. **4**). Thus, when the pilot valve **26** is closed, the flush water flowing into the pressure chamber **20d** from the bleed hole **20c** provided in the main valve body **20a** of the ball tap **16** cannot flow out, and the pressure in the pressure chamber **20d** increases. Then, at time t_6 in FIG. **8**, the main valve body **20a** is pressed by the pressure in the pressure chamber **20d** and seated on the valve seat **20b**, and the main valve body **20a** is closed. This stops water supply from the water supply source **6** via the ball tap **16** to the hydraulic drive mechanism **18** and stops the supply of flush water into the second tank part **15**.

[0087] Upon stopping the water supply to the hydraulic drive mechanism **18**, the piston **18b** (FIG. **5**) in the cylinder **18a** pushed up by the water supply is pushed down by the biasing force of the spring **18c**. The rod **28** attached to the piston **18b** accordingly lowers. The rod **28** lowers to a predetermined position, and the rod **28** is recoupled to the valve shaft **10a** of the jet spouting discharge valve **10** by the clutch mechanism **30**. As described above, toilet flushing is ended once, and the flush toilet apparatus **1** returns to the standby state for the toilet flushing.

[0088] The flush water tank device **4** of the embodiment of the present invention is configured so that the first tank part **13** and the second tank part **15** have different cross-sectional areas A_1 and A_2 depending on the height, and the ratio of the cross-sectional area of the first tank part **13** to that of the second tank part **15** at the first height H_1 (A_1/A_2 in FIG. **6**) is different from the ratio of the cross-sectional area at the second height H_2 that is lower than the first height H_1 (A_1/A_2 in FIG. **6**). For example, as in the above-described embodiment, when the second tank part **15** is configured so that cross-sectional area A_2 is large at the first height H_1 and small at the second height H_2 , a large amount of flush water can be spouted from the jet spout port **2d** at a high hydraulic head pressure in the initial stage of flushing. That is, it is possible to distribute the flush water in the flush water tank main body **12** to the rim spout port **2c** and the jet spout port **2d** in the flush toilet main body **2** in accordance with the required amount of flush water and the water force.

Consequently, the volume in the flush water tank main body **12** and the hydraulic head pressure of the flush water in the flush water tank main body **12** can be fully utilized for toilet flushing.

[0089] Furthermore, according to the flush water tank device **4** of the present embodiment, since the second tank part **15** is configured to have the cross-sectional area that is equal to or more than the cross-sectional area A_1 of the first tank part **13** at the first height H_1 , in the initial stage of

flushing in which the water level in the second tank part **15** is high, the instantaneous flow rate of flush water spouted from the jet spout port **2d** of the flush toilet main body can be increased, and flushing effect can be enhanced.

[0090] According to the flush water tank device **4** of the present embodiment, the second tank part **15** is configured to have the reduced cross-sectional area **A2** below the stepped portion **14e**, while the first tank part **13** is configured to have the enlarged cross-sectional area **A1** below the stepped portion **14e**. As a result, in the initial stage of flushing, the instantaneous flow rate from the jet spout port **2d** can be increased, and concurrently, a sufficient volume can be acquired in the first tank part **13** at the height that is equal to or less than the height of the stepped portion **14e**, so that a sufficient amount of flush water can be allocated to the rim spout port **2c**.

[0091] According to the flush water tank device **4** of the present embodiment, the inner tank **14** is disposed in the flush water tank main body **12**, the second tank part **15** is formed inside the inner tank **14**, and the first tank part **13** is formed inside the flush water tank main body **12** and outside the inner tank **14**. Therefore, the cross-sectional area of each of the first tank part **13** and the second tank part **15** at each height can be freely set, and the amount of flush water and the hydraulic head pressure can be freely set as required.

[0092] According to the flush water tank device **4** of the present embodiment, since the height of the stepped portion **14e** in which the cross-sectional area **A2** of the second tank part **15** is reduced is set to be lower than the jet dead water level **JWL**, the flow rate of flush water spouted from the jet spout port **2d** can be maintained to be sufficiently high until the spouting is ended.

[0093] According to the flush water tank device **4** of the present embodiment, since the stepped portion **14e** in which the cross-sectional area **A1** of the first tank part **13** enlarges is provided at a position higher than the rim dead water level **RWL**, the flush water stored below the stepped portion **14e** of the first tank part **13** can be spouted from the rim spout port **2c**, and a sufficient amount of flush water spouted from the rim spout port **2c** can be acquired.

[0094] Furthermore, according to the flush water tank device **4** of the present embodiment, since the first tank part **13** and the second tank part **15** are configured so that the ratio of the cross-sectional area (**A1/A2**) is reversed at the first height **H1** and the second height **H2**, the instantaneous flow rate of flush water spouted from the jet spout port **2d** can be increased. Concurrently, enough flush water spouted from the rim spout port **2c** can be acquired.

[0095] As described above, the flush water tank device of the embodiment of the present invention and the flush toilet apparatus including the same have been described, but various changes can be made to the above-described embodiment. In the above-described embodiment, the first tank part and the second tank part are formed by disposing the inner tank in the flush water tank main body, and the first tank part and the second tank part may be formed by providing a partition wall in the flush water tank main body.

[0096] Furthermore, in the above-described embodiment, the horizontal cross-sectional area of the first tank part and the second tank part rapidly changes with the stepped portion provided in the inner tank, and the partition wall in the flush water tank main body or the inner tank may be configured so that the horizontal cross-sectional area of the first tank part and the second tank part changes gently depending on the height.

[0097] Furthermore, in the above-described embodiment, the ball tap is used as the water supply valve for switching the supply and stop of the flush water to be stored in the flush water tank main body and the inner tank, and the water supply valve other than the ball tap may be used.

Furthermore, in the above-described embodiment, the jet spouting discharge valve is driven by the hydraulic drive mechanism, and the hydraulic drive mechanism may not be provided.

REFERENCE SIGNS LIST

[0098] **1** flush toilet apparatus [0099] **2** flush toilet main body [0100] **2a** bowl [0101] **2b** discharge trap conduit [0102] **2c** rim spout port [0103] **2d** jet spout port [0104] **2e** rim conduit [0105] **2f** jet conduit [0106] **4** flush water tank device [0107] **4a** lever handle [0108] **6** water supply source

[0109] **8** rim spouting discharge valve (rim spouting switch mechanism) [0110] **8a** bead chain [0111] **10** jet spouting discharge valve [0112] **10a** valve shaft [0113] **12** flush water tank main body [0114] **12a** first drain port [0115] **12b** second drain port [0116] **13** first tank part [0117] **14** inner tank [0118] **14a** drain port [0119] **14b** drain port forming member [0120] **14c** cutout [0121] **14d** cutout [0122] **14e** stepped portion [0123] **15** second tank part [0124] **16** ball tap (water supply valve) [0125] **16a** inflow pipe [0126] **16b** outflow pipe [0127] **18** hydraulic drive mechanism [0128] **18a** cylinder [0129] **18b** piston [0130] **18c** spring [0131] **20** main body [0132] **20a** main valve body [0133] **20b** valve seat [0134] **20c** bleed hole [0135] **20d** pressure chamber [0136] **20e** pressure passage [0137] **22** float [0138] **24** arm [0139] **24a** support shaft [0140] **26** pilot valve [0141] **26a** pilot valve port [0142] **28** rod [0143] **30** clutch mechanism [0144] **32** water supply pipe [0145] **32a** water supply port

Claims

1. A flush water tank device that is attached to a flush toilet main body for use to supply flush water, comprising: a flush water tank main body including a first tank part and a second tank part that store flush water supplied to the flush toilet main body, a rim spouting switch mechanism that switches discharge and stop of flush water in the first tank part so that the flush water in the first tank part is spouted from a rim spout port of the flush toilet main body, and a jet spouting discharge valve that switches discharge and stop of flush water in the second tank part so that the flush water in the second tank part is spouted from a jet spout port of the flush toilet main body, wherein the first tank part and the second tank part are configured to have different cross-sectional areas depending on a height, and a ratio of the cross-sectional area of the first tank part to the cross-sectional area of the second tank part at a first height is different from a ratio of the cross-sectional area at a second height that is lower than the first height.
2. The flush water tank device according to claim 1, wherein the second tank part is configured to have a cross-sectional area that is equal to or more than the cross-sectional area of the first tank part at the first height.
3. The flush water tank device according to claim 2, wherein the second tank part includes a stepped portion provided at a height between the first height and the second height, and the second tank part is configured to have a reduced cross-sectional area below the stepped portion, while the first tank part is configured to have an enlarged cross-sectional area below the stepped portion.
4. The flush water tank device according to claim 2, wherein an inner tank is disposed in the flush water tank main body, the second tank part is formed inside the inner tank, and the first tank part is formed inside the flush water tank main body and outside the inner tank.
5. The flush water tank device according to claim 3, wherein the jet spouting discharge valve is configured to discharge the flush water in the second tank part so that a water level in the second tank part lowers from a predetermined initial water level to a predetermined jet dead water level, and the stepped portion is provided at a position that is lower than the jet dead water level.
6. The flush water tank device according to claim 3, wherein the rim spouting switch mechanism is configured to discharge the flush water in the first tank part so that the water level in the first tank part lowers from the predetermined initial water level to a predetermined rim dead water level, and the stepped portion is provided at a position that is higher than the rim dead water level.
7. The flush water tank device according to claim 1, wherein the first tank part and the second tank part are configured so that the ratio of the cross-sectional area is reversed at the first height and the second height.
8. A flush toilet apparatus that is flushed with flush water stored in a flush water tank, comprising: a flush toilet main body including a bowl, a rim spout port, and a jet spout port, and the flush water

tank device according to claim 1, which supplies flush water to the rim spout port and the jet spout port of the flush toilet main body.
